Jingkai LAN

Curriculum Vitae
bioljk@outlook.com - lanrem-ljk.github.io

Education

M.Sc. Bioinformatics and Computational Biology Sep 2023 - Jan 2026
University of Bern, Switzerland
GPA: 5.0 / 6.0 (magna cum laude)

Summer School: Machine Learning Jun 2024
University of Bern, Switzerland

B.Sc. Biological Science Sep 2019 - Jun 2023
Shihezi University, China
GPA: 3.64 / 4.0

Research Experience

Advancing Metal-Binding Protein Predictions with Deep Learning Feb 2025 - Sep 2025
Thomas Lemmin Lab, IBMM, University of Bern - Master's Thesis

- Developed a deep learning framework to predict metal-binding proteins and enzymatic properties from amino acid sequences alone, reducing the cost of experimental annotation.
- Designed a data augmentation strategy using in silico mutagenesis at metal-binding sites (ProteinMPNN), with quality control via structure prediction tools (CHAI-1, ESM).
- Trained protein language models (ProtBERT, ESM-2) and embedding-based classifiers (XGBoost) under supervised and semi-supervised paradigms.
- Achieved performance on par with leading approaches that leverage structural information, without requiring extensive feature engineering.

Tools: ProtBERT, ESM-2, GANBERT, ProteinMPNN, CHAI-1, XGBoost, MMseqs2, ProDy, TMalign.

NeurIPS - Open Polymer Prediction 2025 Jun 2025 - Sep 2025
University of Notre Dame, NeurIPS 2025 Competition Track

- Built machine learning pipelines to predict polymer properties from SMILES-based molecular representations.
- Extracted molecular graph features and fingerprints with RDKit; applied XGBoost, LightGBM, and ChemBERT with systematic hyperparameter optimization.
- Ranked within the top third of the final leaderboard.

Tools: SMILES, RDKit, ChemBERT, XGBoost, LightGBM.

Diversity of Peach Tree Viruses in Xinjiang, China Nov 2021 - Aug 2022
Baiming Cui Lab, Shihezi University - Bachelor's Thesis

- Performed bioinformatics analysis of metatranscriptomic data from Prunus persica var. compressa to identify viral species, profile abundance patterns, and characterize genetic relationships.
- Conducted phylogenetic analyses to investigate genetic diversity and evolutionary status of peach-associated viruses in the Xinjiang region.

Tools: Python, Megahit, Bowtie2, BLAST+, MEGA7, Samtools, Trimmomatic.

Morphological and Molecular Survey of Bat Species Nov 2020 - Aug 2021
Abduriyim Shamshidin Lab, Shihezi University - Research Training

- Collected and morphologically identified naturally deceased bats in the Shihezi region (Xinjiang).
- Extracted genomic DNA, amplified mitochondrial ND1 and Cyt b genes via PCR, and confirmed species identity as Pipistrellus pipistrellus through sequence alignment.
- Provided the first combined morphological and molecular evidence for the species' presence in the region.

Tools: PCR, DNA Extraction, NCBI BLAST, Sequence Alignment, Morphological Identification.

Skills

Programming & Computing: Python, R, Bash, High-Performance Computing (HPC)
Machine Learning & AI: Deep Learning, Machine Learning, Protein Language Models, Gradient Boosting (XGBoost, LightGBM), Semi-supervised Learning
Bioinformatics: NGS Analysis, Sequence Alignment, Phylogenetics, Metatranscriptomics, Molecular Feature Engineering (SMILES, RDKit)
Laboratory: Nucleic Acid Extraction, Gel Electrophoresis, PCR, Spectrophotometer
Languages: Chinese (Native), English (B2-C1)

Awards and Honours

Excellent Graduate (one of 17 among 180 students) Dec 2022
Summer Camp Excellent Participant Award, ShanghaiTech University Jul 2022
First-Class Scholarship (one of 5 recipients among 104 students) Dec 2021